DO NOW — RECALL

Run-length and dictionary coding

Name: _____ Class: _____ Date: _____

1. Define lossless compression.

2. What is a run in RLE?

3. What is a compression overhead?

4. For array [2,3,4,5], double values at indexes 1 and 3. State final array and sum.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Original can be reconstructed exactly.

2. Model answer

Consecutive repeated instances of the same symbol.

3. Model answer

Extra metadata needed to represent/decode data.

4. Model answer

[2,6,4,10], sum 22; indexes are positions 1 and 3.

Vocabulary — know these terms

RLE — Encode repeated runs with counts.

run — Maximal consecutive equal symbols.

dictionary coding — Represent repeated strings by references.

dictionary — Mapping of codes to values/phrases.

overhead — Metadata/structure required by encoding.

NEW LEARNING

Read, annotate and explain

Understand

Run-length encoding replaces each maximal run of the same symbol with a value and count; it works best for long consecutive repetitions.

Apply

Dictionary coding replaces repeated strings with dictionary references; the decoder needs the same dictionary, transmitted or reconstructed according to the scheme.

Evaluate

Overhead matters. Alternating characters can make naive RLE larger; long repeated phrases can be valuable for dictionary coding.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Encode BBBBAAC as symbol/count pairs and verify decoding.

Predict first

Identify maximal consecutive runs: four B, two A and one C.

Model and check

Output each run as value/count and expand them back to verify exact reconstruction.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

RLE-encode AAAAABBCCCDDDD with counts. [4]

Question 2

With dictionary 0='HELLO',1='WORLD', encode 'HELLO WORLD HELLO' using references and separators. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain why RLE can increase size for alternating data and why dictionary coding may help repeated long substrings. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

RLE always shortens any input file.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.